EFY TESTED
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WATER PUMP CONTROL
For Rented Houses

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In a row of rented houses that share a common pump motor for their water supply, issues arise when it comes to splitting the electricity bill. The shared motor often runs on a common main supply, and it is challenging to divide the electrical charges properly among tenants due to variations in water consumption.

The control circuit presented here offers a solution by allowing the common pump motor to run on the supply specific to each house whenever a tenant switches on the motor for their water needs. To achieve this, each house is equipped with individual distribution discharge lines with manual valves that connect to their overhead water tanks. Before starting the pump to fill their overhead tank, a tenant must ensure that their respective discharge valve is open while the others' valves are closed.

This innovative circuit enables the pump to run on power from different main supplies, even accommodating different phases in a three-phase system. The setup enables more accurate billing of electricity consumption based on individual usage. Fig. 1 showcases the author's prototype and the Bill of Materials table lists the components required to build this project.

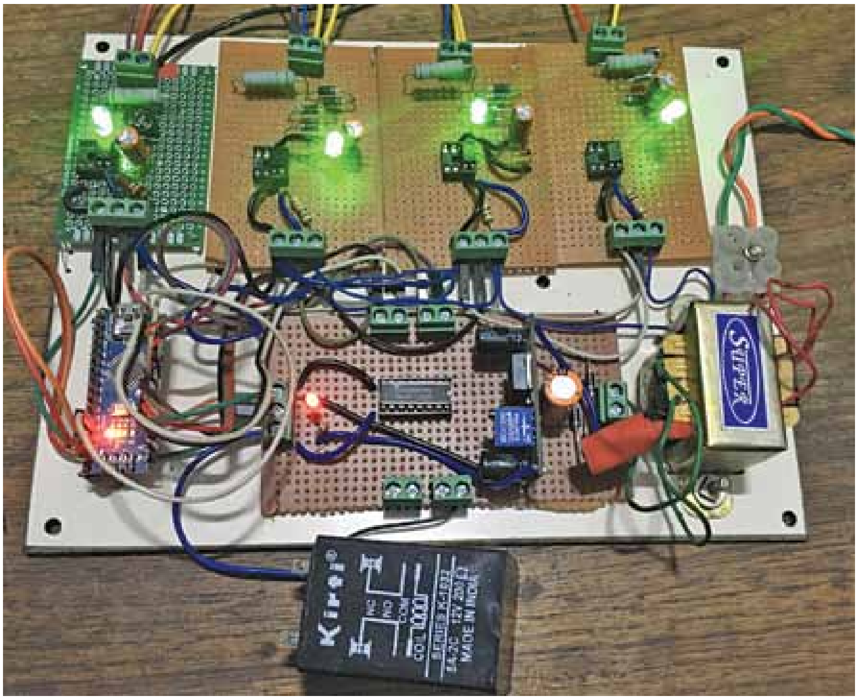

Fig. 1: Author's prototype

Circuit and working

The circuit diagram of the water pump control for rented houses, shown in Fig. 2, is based on an Arduino Nano board (MOD1), four PC817 optocouplers (IC1 through IC4), relay driver IC ULN2003 (IC5), a 12V voltage regulator (IC6), and various other components.

The circuit can be divided into two sections: the power circuit and the control circuit. The power circuit includes the pump, fuses, zener diodes, diodes, and more. It consists of the pump (motor) power supply receiving terminal blocks from each house, motor on/off power switches, power fuses (F1 through F4) for short-circuit protection (miniature circuit breakers MCB of equal ratings can be used), motor switching relays (RL1 through RL4), and

BILL OF MATERIALS

Components	Description	Quantity
Arduino nano board	MCU	1
IC-7812 (IC6)	12V voltage regulator	1
ULN2003 (IC5)	Relay driver	1
PC817 (IC1-IC4)	Optocoupler	4
FZ5.1A (D1-D4)	Zener diode	4
1N4007 (D5, D6-D16)	Diode	16
LED (LED1-LED4	5 mm LED	4
Resistor (R5-R8)	10K, 1/4W	4
Resistor (R1-R4)	15K, 2 W	4
Capacitor (C1-C4)	100uF, 25V electrolytic	4
Capacitor (C5, C6)	0.01uF, ceramic disc	2
Relays (RL1-RL4)	12V DC, 6A, 2 C/O	4
Glassfuses (F1-F4)	250V/500V, 16A	4
AC to DC power adaptor	15V, 1A Output	1
Power terminal block strip	(12 Ways)	1